

Wayfinder: An Independent On-Device Recommender Algorithm for YouTube

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We present Wayfinder, a browser extension that enables users to run an on-device independent YouTube recommender algorithm, powered by an experimental content querying API. It gives users freedom over their online experience, which we believe is the only realistic way to tackle the problems caused by the algorithm. We believe large platforms like YouTube should be required to expose the APIs needed for independent recommender systems to exist, opening up genuine competition in the recommender algorithm space.

Motivation

YouTube is a central part of modern life, especially for young people. [90% of US teens](#) use YouTube, with 73% watching it daily and 15% watching it “almost constantly”. YouTube’s recommender algorithm has [significant influence](#) over the content shown to users. The engineers designing these algorithms are incentivised to maximise time spent on the platform at the expense of user wellbeing. YouTube’s recommender algorithm has been accused of causing various harms, such as [encouraging self-harm](#) and [spreading misinformation](#). These recommender systems are also highly privacy-invasive, as they collect a large amount of [detailed personal information](#) in order to make recommendations.

We believe that users should have full control over the content that is recommended to them on the internet. We built Wayfinder to prove that it is technically feasible to separate the recommender algorithm from the underlying platform, such that the algorithm can be designed to maximise user wellbeing instead of platform profits. We believe this is the only realistic way to break out of the attention economy, as first-party recommenders will always be incentivised to maximise time spent on the platform above all else.

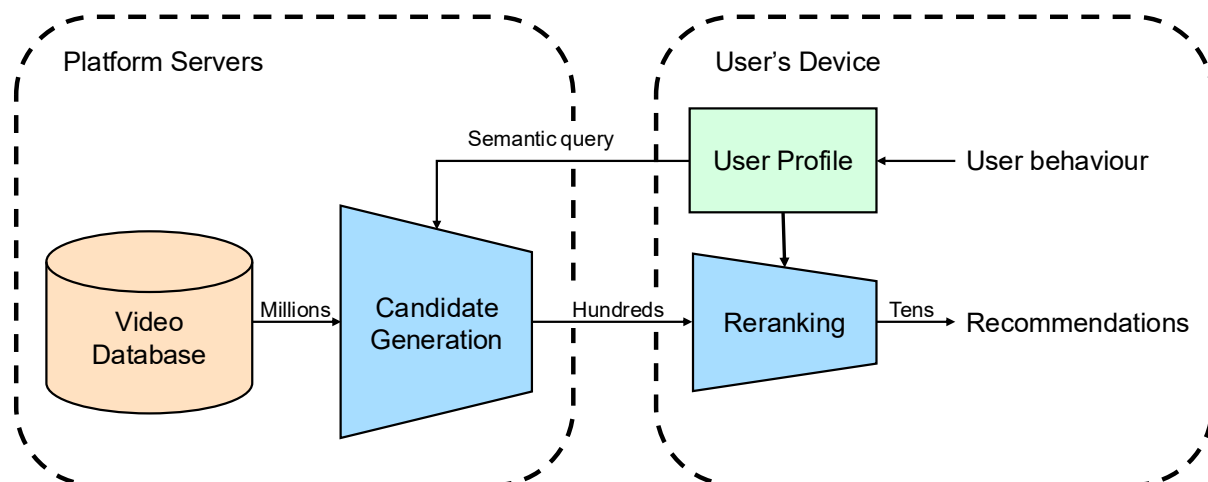
How Recommender Algorithms Work

Most recommender systems follow a [similar structure](#). They are split into two steps: candidate generation and reranking. Candidate generation retrieves hundreds of candidate items from the content database based on some coarse query signals. Reranking then sorts these retrieved items based on a more sophisticated scoring regime, to select the top few items to actually show to the user.

In most recommender systems, both of these parts live on the platform provider’s servers, so the platform has full control over how they operate. The goal of Wayfinder is to move the “opinionated” parts of the system onto the user’s device or a third-party service, giving users control over the algorithm logic instead of the platform owners.

Candidate generation inherently requires access to the entire content database, so it must live on the platform servers. Reranking on the other hand only needs to operate on a few hundred videos returned by the candidate generation phase, so it can be trivially moved to the user's device. Reranking is where most of the fine-tuned algorithm logic lives, so moving it to the user's device gives the user a great deal of control over the behaviour of the system.

Wayfinder also aims to give users control over the candidate generation phase by exposing a semantic query API that the user's on-device algorithm uses to control the candidate retrieval process, even though it runs on the platform's server. The user's recommender system tells the platform servers exactly what to retrieve, with minimal room for the platform server to influence the returned results to their own ends. This setup also minimises the amount of personal data sent to the platform, as only small queries are sent to the servers, instead of the user's entire behavioural profile, as would traditionally be the case.

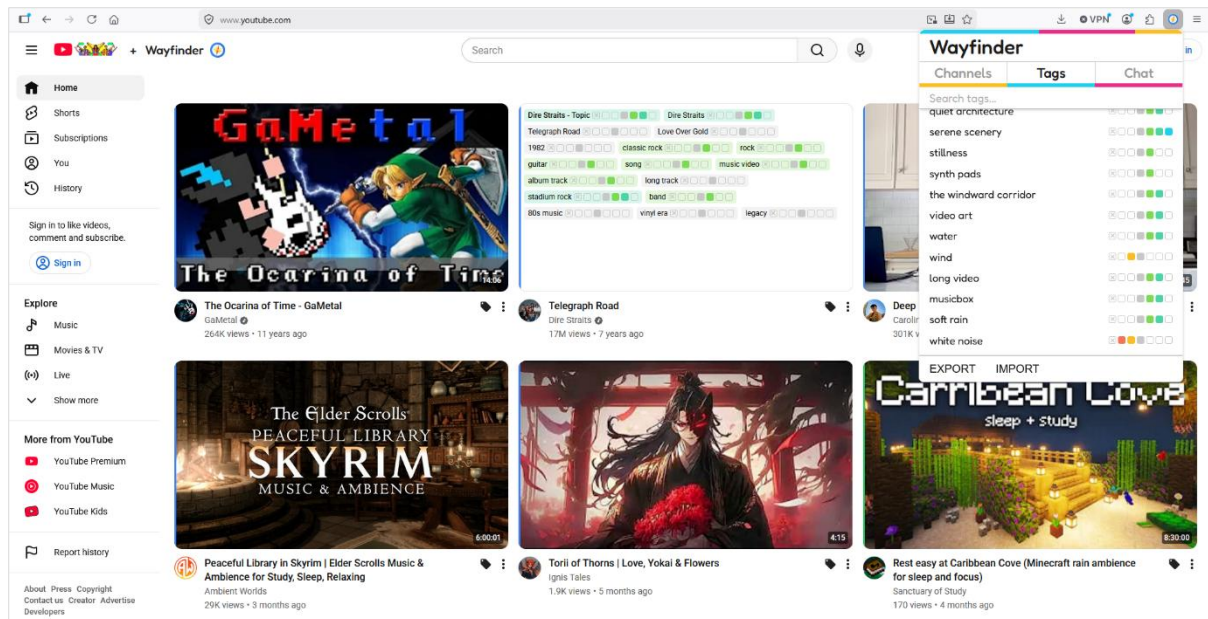


Wayfinder's system architecture. In a typical recommender system, all these components would live on the platform servers.

How Wayfinder Works

Wayfinder generates tags for all videos based on their title and description. As a user watches videos, tag scores are inferred based on their behaviour and stored on device. These tag scores form the user's profile. When Wayfinder needs to generate recommendations, it sends a subset of these tag scores to the video index server to retrieve a large number of relevant candidate videos that match those tags. Local reranking logic then sorts these candidates based on things like prior watch time, variety and so on.

Unlike YouTube, Wayfinder allows users to view their entire tag profile, and export and import it from a local JSON file for total control. Users can manually tweak the scores of tags associated with a video on the watch page, making it easy to alter the behaviour of the system. Wayfinder also has an AI chat feature that allows users to directly tell the recommender algorithm what they want to see, by proposing tag score changes based on a prompt.



A screenshot of Wayfinder on the YouTube homepage. Users can inspect the tags used to recommend a video, and edit their entire score profile.

Proposed Regulation

Wayfinder demonstrates that independent, on-device recommender algorithms are technically feasible and bring real benefits to end users. We call on regulators to mandate that large platforms provide the APIs necessary to operate systems like Wayfinder at scale, opening up genuine competition in the recommender algorithm space. We want regulation to mandate the following two things:

1. A semantic query API that third-party services can use to retrieve candidate videos based on attributes of the videos. These attributes must have a human-observable correlation to the content of the video, such that inaccurate labels can be contested.
2. Settings in the YouTube app to allow users to select a third-party recommender system. Third-party recommender systems must have access to all the signals that the first-party recommender does. First-party recommenders must only retrieve candidate videos via the public API. Advance warning must be given to change any of the interfaces.

Ideally wider vertical interoperability would allow more innovation in third-party user interfaces that enable greater user control and visibility over third-party recommenders, but integration into the first-party YouTube app is the bare minimum requirement for now.